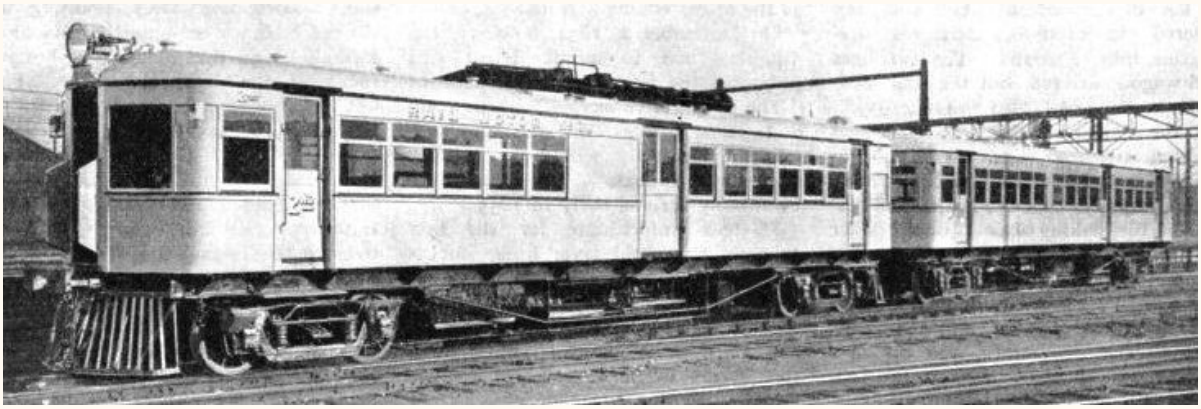
Across Australia by Train

Pioneers of an Unparalleled Wilderness

RAILWAYS OF THE COMMONWEALTH - 4



MODERN TRANSPORT METHODS are a feature of the Australian railways. This picture shows a Rail Motor on the New South Wales system.

THE modern traveller arriving in Australia finds trains as distinctive in the railway world as the kangaroo is in the animal world, for the island continent is the land that is different from other countries.

There are trains which, like the camel, have to carry a hump of extra water to cross a wilderness unparalleled in both hemispheres; “poison” trains to kill weeds that would choke the track if not dealt with promptly; sheep trains which are double-deck service flats for the gentlemen who pay Australia’s rent; trains which climb heights equal to Snowdon; trains which take city men into the back-blocks to show them the resources of the country, and trains which take agricultural experts from the centres of government into the pastoral areas to teach farmers the latest methods in stock-raising and farming.

There are railroads which have been held up because all the navvies have dashed off to search for gold; railroads built specially to enable men to hurry to the goldfields; railroads which handle rush-hour traffic of a density that would make a London Transport official ponder; railroads where passengers are so rare that in a whole year one station took only five-pence from the solitary passenger who used it.

There is the 3,384-miles link between Fremantle, the port for Perth, on the Indian Ocean side of the continent, to Brisbane, on the Pacific side, to contrast with short tracks which tap the sugar-cane fields of Queensland.

In considering Australian railroads it is important to remember that in the middle of the last century the continent was isolated. Australia has now been reached in a week-end dash by air from England. There are people alive to-day who can remember that a trip to Australia took many weary months. In fact, sail was faster than steam when the railway was laid down in New South Wales and Victoria. For example, the Black Ball Line clipper, *Marco Polo*, reached Melbourne from Liver-pool in seventy-eight days, arriving on July 4, 1852, and was followed by *Lightning*, a super-clipper. For years these fast sailing ships defied the steamship.

But Australia was determined to have railways, and the experience of the Melbourne and Hobson’s Bay Railway Company, is a typical example of her determination. The company ordered locomotives, carriages and wagons from England. The carriages and wagons arrived, but the four locomotives did not. This was a set-back, but the

directors wanted to go ahead without delay, so they ordered a locomotive from Robertson, Martin, Smith and Company, of Melbourne, and entrusted the making of parts of the boiler to the Langlands Foundry. A 30-hp locomotive was completed in ten weeks, and was the first one built in Australia.



THE RAILWAY LINK from Fremantle, on the western side of Australia, to Brisbane, the capital of Queensland.

Thus on September 12, 1854, the first railway in Victoria and the first locomotive-operated railway in Australia was opened. The gauge was 5 ft 3-in and the distance was over two miles between Flinders Street, Melbourne, and Port Melbourne.

On the opening run the engine achieved a speed of fifteen miles an hour, but bad fuel on September 16 and damage to the locomotive on September 18 hampered the running, and for a while trains were discontinued. But the directors, not wanting the line to be idle while the engine was being repaired, remembered the existence of a pile-driving engine fitted to a truck which had been used for drawing wagons during the building of the railway. This was brought into use and pulled the passenger trains as quickly as the larger engine had done.

On December 2, 1854, however, the directors had to admit defeat and published the following advertisement: “The Loco-motive having broken down, and those from England not yet landed, the train will cease running until further notice.” The locomotives eventually arrived early in 1855.

It was unfortunate for the first railways that gold fever broke out as the lines were being laid. Workers left for the gold fields and costs soared like rockets. The little line between Sydney and Parramatta, fourteen miles long, took years to build and cost £50,000 a mile. At one time 500 navvies imported by the company deserted. In Adelaide labourers demanded one pound a day, a driver with a horse and cart demanded twelve pounds a week.

The Quest for Gold

Gold had been discovered earlier, but it had been kept a secret. In 1839 a Polish Count, Paul Strzelecki, on a scientific exploring expedition from Sydney across the mountains into the region of Victoria which he called Gippsland, observed particles of gold among decomposed iron-stone. The maps and specimens were examined in England by Sir Roderick Murchison, who wrote to the Secretary of State for the Colonies, Lord Grey, pointing out the resemblance between the geological formation and that of the gold-bearing rocks of the Ural Mountains. This letter was ignored.

In 1823 a surveyor had picked up some specks of the precious metal. Years later a Sydney geologist, W. B Clarke, made observations in the Bathurst region and showed a specimen to Governor Gipps. Gipps, dreading the unrest which the news would cause among the convicts, said to the geologist: “Put it away, Mr. Clarke, or we shall all have our throats cut.” For this reason the Governor asked Count Strzelecki to say nothing and the Count complied, omitting the subject in his first book on Australia. In 1848, a piece of gold found near Berrima was shown to the officials in Sydney, but they would not order a survey.

In many places, however, men were finding gold and the secret could not be kept. A shepherd found gold at the roots of a tree which had been blown down; a labourer digging a hole for a fencing post struck a nugget with his spade and sold it for a hundred pounds: these men talked.



BRISBANE CENTRAL STATION. From this terminus in the capital of Queensland, lines radiate in all directions. There are over 6,400 miles of railway track in Queensland, most of the lines having a gauge of 3 ft 8-in.

Edward Hargreaves, who had been a sheep farmer on the Bathurst Plains, went to the Californian goldfields; he saw the similarity in the formation of the rocks and hurried back to Australia. In vain did the Lands Commissioner at Bathurst write to Sydney in 1851 complaining that Hargreaves was employing men to dig for gold. Hargreaves had already written to Governor Fitzroy, saying that he could point out where gold was to be found if he were rewarded; eventually he was given £10,000.

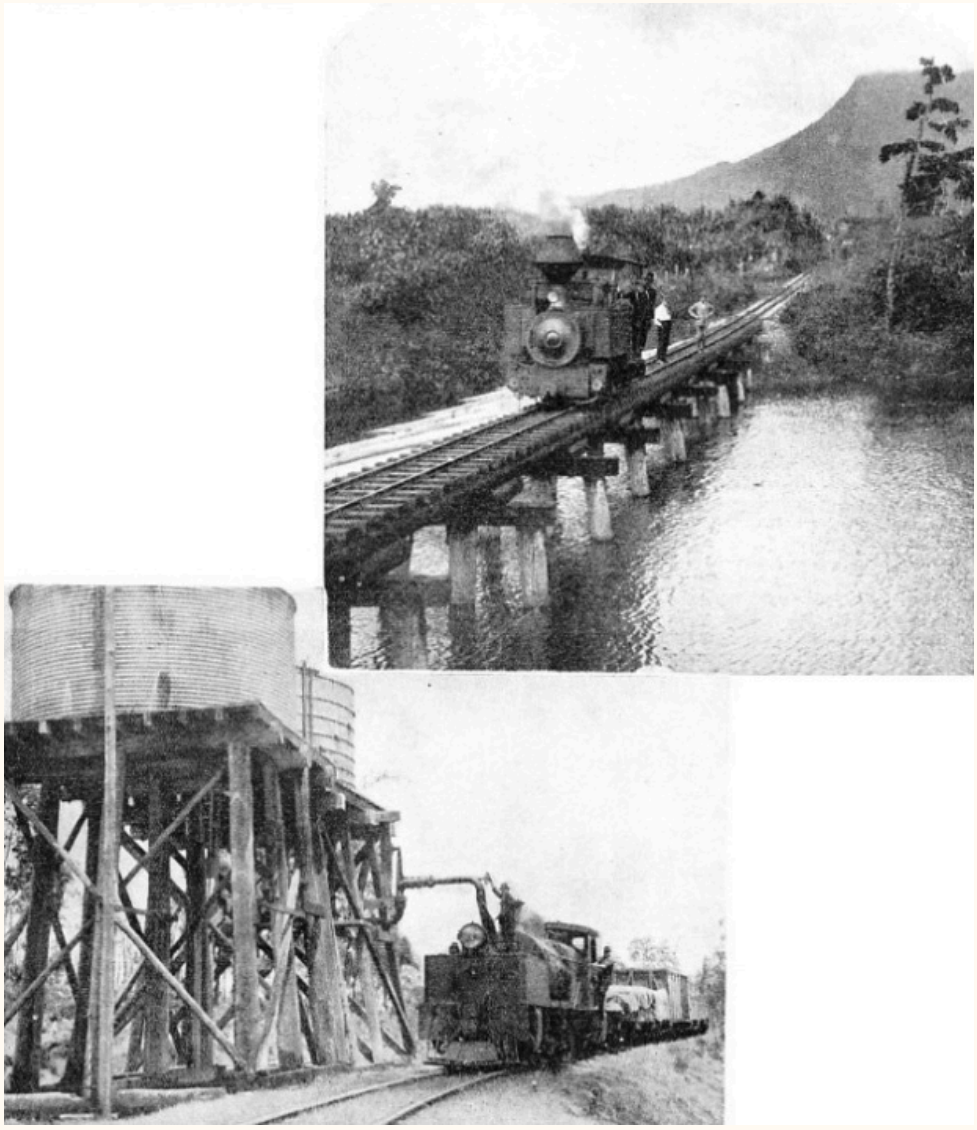
Directly the news came to Melbourne the gold rush began. Month by month fresh strikes were made and Ballarat and Bendigo were tapped; before the end of the year gold worth nearly £900,000 had been found in Victoria. From every country men poured into Australia, and if a man like Lord Robert Cecil, afterwards Marquis of Salisbury, Prime Minister of Great Britain, joined in the rush it is not to be wondered at that the labourers imported to construct the railways did like-wise. The financial throats of the shareholders in the new method of transport were, metaphorically, cut. Time and again the companies had to be helped by the Government.

A SPECIAL LOCOMOTIVE used for sugar-cane traffic, crossing the Russell River in Queensland.

REPLENISHING THE TANK of a Garratt locomotive attached to a freight train at Brookdale, in the southwest timber district of Western Australia. Jarrah, the timber, is famous for its resistance to decay and insects, and is used for railway sleepers and for building.

The Sydney Railway Company began work on a line to Parramatta on July 3, 1850, but the gold rush intervened, and it was not until September 26, 1855, that the railway was opened for traffic.

This meant that it had taken more than five years to build fourteen miles. In Melbourne the first locomotive had made its run on a two-miles track just over a year before. In Adelaide, in 1856, the first State-owned steam railway - seven miles in length - in the British Empire became operative between the City and Port Adelaide. A few ambitious pioneers in Adelaide had urged the Government to action, demanding to know “why George



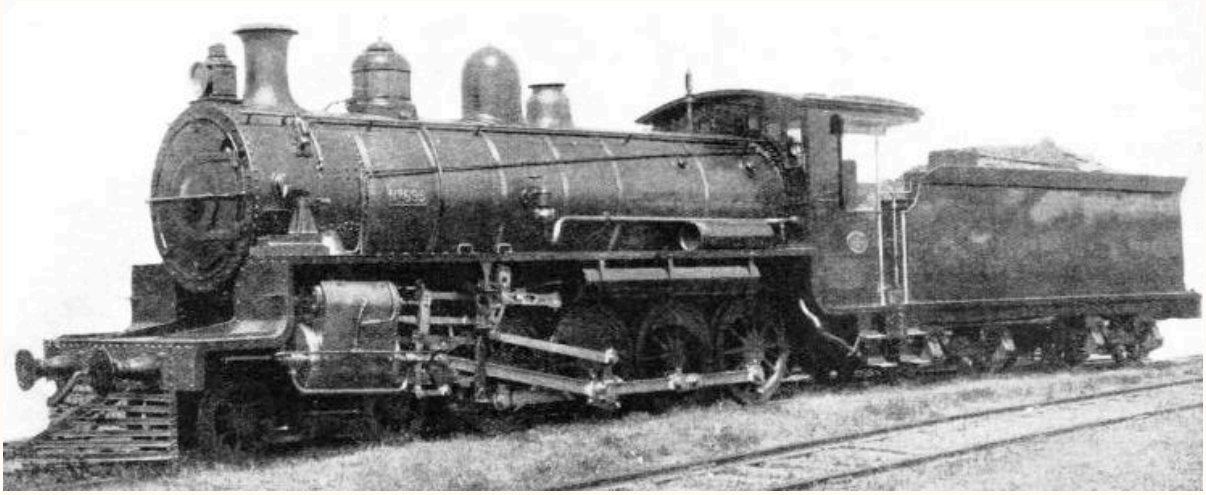
Stephenson should be supplying locomotive traction to Englishmen while South Australians were left to wield the bullock whip, taxing both their strength and their vocabulary?”

In those days the citizens of Adelaide had to pay £3 a ton for freight cartage by bullock wagon for those seven miles. The railway cut the rate down to 4s 6d a ton.

Before the State Railway was opened the people of Adelaide had to walk to Port Adelaide, unless they could pay for a ride in a bullock dray or for a lift in an English gig that bumped over the road; they were so impatient that in 1854, before the locomotives arrived, a railway was already in operation between Goolwa and Port Elliot, but the “locomotives” were horses.

Fares on the State Railway between Adelaide and Port Adelaide were 2s 6d first-class, 1s 6d second, and 1s third. (There is now no third-class in Australia.) These fares compare favourably with those ruling previously when a spring van did the journey each day, Sunday excepted, at a fare of 4s.

But whereas the tracks in Melbourne and Adelaide were of 5 ft 3-in gauge, those in Sydney were 4 ft 8½-in. The Australian gauge muddle had begun.



FOR GOODS TRAFFIC on the Queensland Government Railways. This locomotive has a 4-8-0 wheel arrangement, a tractive effort of 22,141 lb, and a boiler pressure of 160 lb. The diameter of the driving wheels is 48-in, and that of the cylinders 18 in.

This matter must be approached with caution. Australians are very sensitive about it. In 1846, Mr. Gladstone, while Colonial Secretary, recommended the 4 ft 8½-in gauge for Australia, but in 1852 New South Wales appointed an Irishman as chief engineer. He had been used to the 5 ft 3-in gauge in Ireland, and New South Wales took his advice, despite what Mr. Gladstone had said. Victoria and South Australia began to build; they had been awaiting the decision of New South Wales. Then New South Wales changed their chief engineer from an Irishman to a Scotsman, and he advocated the standard gauge of 4 ft 8½-in. He won the New South Wales Government over and they changed to the standard gauge, repealing the Act they had passed in 1852; in 1855 they adopted the 4 ft 8½-in gauge without consulting Victoria and South Australia. Victoria had already ordered rolling-stock and locomotives. As the most rapid means of communication was by sail, and as it was impossible to get into touch with England to cancel the contracts, the mischief was done.

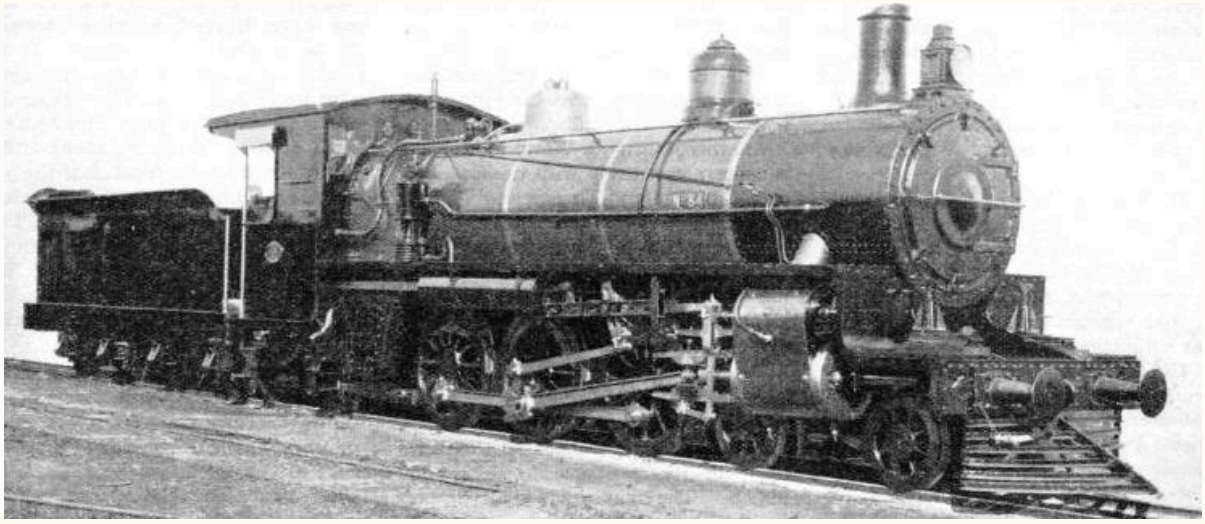
Later on. New South Wales appointed yet a third engineer, who advised that the twenty miles of track laid when he took over the job should be torn up and replaced by broad gauge, but the Government dismissed the possibility of their lines ever meeting those of the Victorian railways as a “very remote contingency”, and the suggestion was rejected.



A LOG TRAIN on a private timber line at Manjimup in Western Australia.

It is easy in these days to get heated about the muddle, but it is more just to visualize the conditions at the time. Nearly a score of years were fated to pass before the deep-sea cable brought Australia and London into rapid communication; also the Australians knew little about steam but much about mountains. The notion that locomotives would be able to pull trains over those mountains was no more than an idea. Just as in more recent years, even in England, men scoffed at the idea of the aeroplane, so these men in an isolated community did not realize that the locomotive could conquer the mountains that had hemmed them in for so many years. They knew the mountains: they did not know the steam locomotive.

That is why, to-day, New South Wales is an “island” of standard gauge, and the only two States connected by lines of an equal gauge (5 ft 3-in) are Victoria and South Australia. In Queensland, to the north of New South Wales, the gauge is 3 ft 6-in, because Queensland found that the narrow gauge is cheaper. In recent years a new line has been laid from Brisbane into New South Wales on the standard gauge, linking Sydney and Brisbane. Victoria and New South Wales are the two States that should have agreed from the beginning, as they contain the bulk of the population and they constitute the economic heart of the Commonwealth. An Irishman and a Scotsman, both eloquent and persuasive, a new country with governments susceptible to eloquence, plus something novel - result: confusion.



THIS 4-6-2 TYPE PASSENGER LOCOMOTIVE is used on the Queensland railways. The engine has a tractive effort of 20,062 lb, and a boiler pressure of 160 lb. The cylinders are of 18¼-in diameter, and the diameter of the driving wheels is 51-in.

It is most unlikely that the muddle will ever be straightened out, for the reason that it would cost too much. Attempts have been made, but it is doubtful if they will succeed. Any reader who approaches the subject of Australian railways with the idea that he will be able to step off a liner at Fremantle, Western Australia, board a train, and step off at Brisbane, Queensland, must remember the foregoing, and be under no illusions: Australia is different, even to the gauges, more than either Europe or America.

The Trans-Australian Railway is a great feat, but the title is misleading to anyone who is not familiar with Australia. In con-sidering Australia as a whole one must remember the Trans-Australian and the New South Wales Government Railways as “islands” of standard gauge separated by narrow and broad gauges.



GRAIN TRANSPORT forms one of the most important items in the revenue of the Australian railways. This illustration shows how trucks are loaded with bags of wheat in Western Australia.

The Sydney Railway Company made Mr. Connell, locomotive superintendent of the London and North Western Railway, their Consulting Engineer, and it was he who took charge in England of the construction and transport of the equipment.

The first engines, of which there were four, were modelled closely on some 0-6-0 engines designed by Connell, and which were then working on the London and North Western Railway. The alterations consisted mainly of replacing the trailing driving wheels by a pony truck, making them 0-4-2 engines. They were built by Robert Stephenson, had cylinders 16-in by 24-in, coupled wheels 5 ft 6-in diameter, a steam pressure of 120 lb, and they weighed 33¾ tons. The tender, a six-wheeler. weighed 22½ tons.

For years the Australian railways imported their locomotives from Great Britain and the United States until their own work-shops developed and supplied the requirements of the Commonwealth.

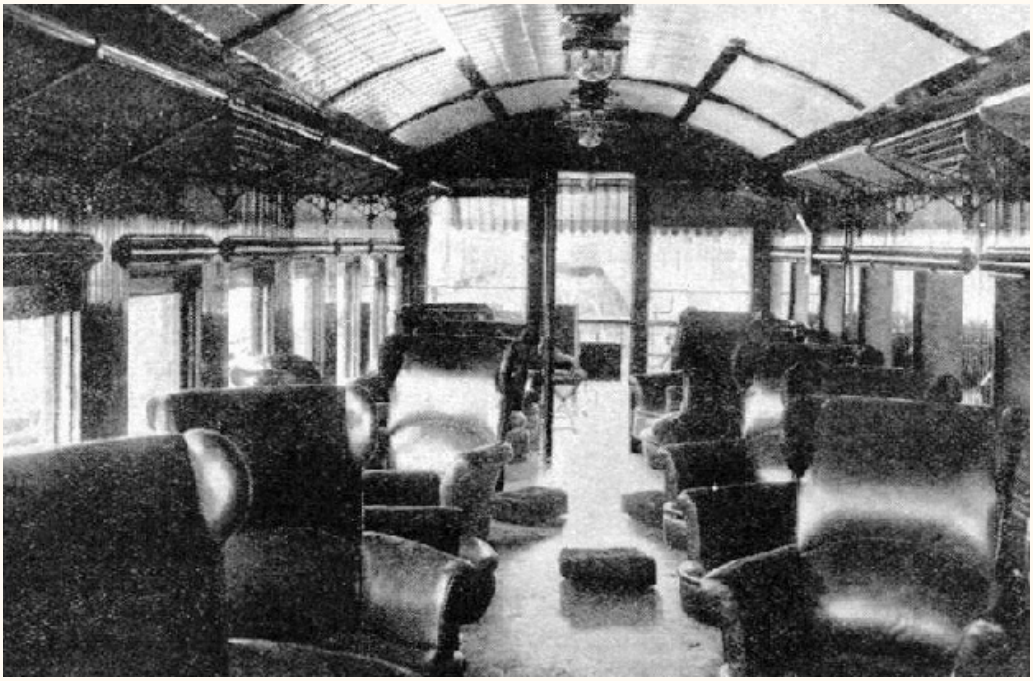
Except for one railway - the Midland Railway of Western Australia - which operates about 277 miles of 3 ft 6-in gauge track from Midland Junction, near Perth, to Walkaway, on the West Coast, there is no company-owned passenger-carrying rail-way of considerable size in the continent. This company is concerned mainly with carrying wheat and general merchandise.

Another railway in Western Australia is concerned entirely with timber. It is owned by Millars' Timber and Trading Company, of Perth, and consists of 239 miles of 3 ft 6-in gauge track. There are eight lines operating in lengths varying from four to sixty-three miles, located in sections of the Darling Ranges along the south-western coast.

£5,000 a Mile

The railway taps the Jarrah forests, and the timber is famous all the world over for its resistance to decay and insects. The trees provided the sleepers for the Trans-Australian Railway.

The gauge of 3 ft 6-in is used in Western Australia, Queensland, part of South Australia and in Central Australia and on the North Australia Railway. It was adopted because of its cheapness, the capital cost in Western Australia being kept down to about £5,000 per mile, a low figure compared with that of other States.



IN QUEENSLAND these attractively furnished parlour cars are attached to long-distance trains. Their light and airy interiors make them especially popular.

The discovery of gold in Western Australia encouraged the growth of railways. The boom began in the 'nineties, when two men made a rich strike at Coolgardie. The towns of Coolgardie and Kalgoorlie sprang up in what had been the desert, and the railway followed the prospectors.

In this State the track laid until 1879 was only thirty-five miles (between Geraldton and Northampton, on the West Coast), and by 1890 the figure crept up to 430. This was due to the building of a track from Fremantle to Perth, on to Northam, and then south to Albany on the south coast, but the gold boom in the next decade sent the lines deeper into the interior, and the total up to 1,632 miles. This was almost doubled by the time the Great War broke out in 1914, and is now, taking no account of the Trans-Australian Railway, nearly 5,000 miles.

The Central Australia Railway, owned by the Commonwealth Government Railways, connects Port Augusta, South Australia, with Alice Springs, in the heart of the continent, by a track 771 miles long. The track crept up by degrees from Oodnadatta. and when, in August, 1929, the first passenger train arrived, the entire population of Alice Springs, numbering about 200, turned out to welcome it. The parlour coach and the dining-car were objects of wonder and admiration. The cost of living came down rapidly, petrol prices were cut by one-half to 3s 2d a gallon, and the cost of transporting groceries, which had been £53 per ton, was considerably reduced.

In considering the subject of Australian railways it must be remembered that the great bulk of the people of Australia live near the south-east coasts. A glance at the map shows that the greater part of the interior is uninhabited. It is, therefore, useless to compare Australian conditions with those obtaining, say, in Canada. For many years the early settlers in New South Wales were prevented by the mountains from spreading westwards, and when a road was cut through the mountains it was later used by the railway pioneers. Isolated as they were, each of the States - New South Wales, Victoria, South Australia, Western Australia and Queensland - developed individually and, particularly between New South Wales and Victoria, a spirit of emulation was engendered. Each State had different laws and varying railway gauges. The Common-wealth did not come into being until January 1, 1901.

"PUFFING BILLY" is the nickname of this timber-train operating near Powelltown, Victoria. It is employed for hauling logs and taking provisions and material to the lumbermen. The smoke-stack is fitted with a spark-arrester - a necessary precaution in the wooded areas through which the train passes.



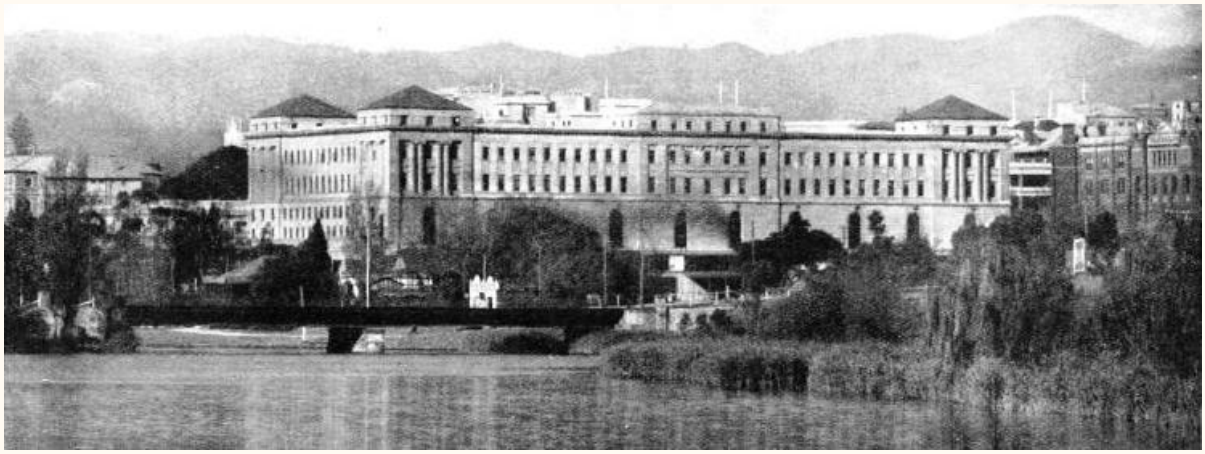
By then four capitals, Brisbane, Sydney, Melbourne, and Adelaide were linked by rail, but Western Australia was cut off by the desert and not until 1917 did the transcontinental railway connect Perth with the other capitals. Indeed, the desert rendered Western Australia an island: the only communication was by sea. The two dead ends at Kalgoorlie, Western Australia, and at Port Augusta, South Australia, were linked by a single track 4 ft 8½-in gauge, making it possible to travel from Perth, the capital of Western Australia, to Brisbane, 3,372 miles.

Owing to the difference in gauges at various places, it is not possible to travel all the way in the same train.

The linking of Kalgoorlie with Port Augusta by the Trans-Australian Railway is an engineering achievement of which the Commonwealth is justly proud. The railways of the Western State had been pushed eastwards to the mining city of Kalgooriie, 375 miles from Perth, and the railways of the State of South Australia had spread north and west from Adelaide to Port Augusta, a distance of 260 miles. Between Kalgoorlie and Port Augusta lay 1,051 miles of practically waterless and uninhabited country. One or two explorers had, in a period of forty years, succeeded, after terrible hardships, in getting across, but from the commercial point of view the barrier was impassable.

The country through which the railway now runs falls into four divisions: (1) The granitic plateau, extending for 167 miles eastward from Kalgoorlie; (2) The limestone or Nullarbor Plain, which runs for 450 miles to the east from the edge of the granitic country; (3) The belt of sand hills on the eastern edge of the limestone region, through which the line runs for about fifty miles; (4) The stretch of country extending for nearly 400 miles from the eastern edge of the sand hills to Port Augusta.

There was not a single running stream in the whole of the distance and probably not a drop of drinkable surface water for about 800 miles of it, while the supplies of water on the remaining 250 miles were almost negligible. For 800 miles there was no settlement of any kind.



ADELAIDE STATION, the headquarters of the South Australian Government Railways. £350,000 was recently spent on it, and this railway station is now one of the finest in the British Empire.

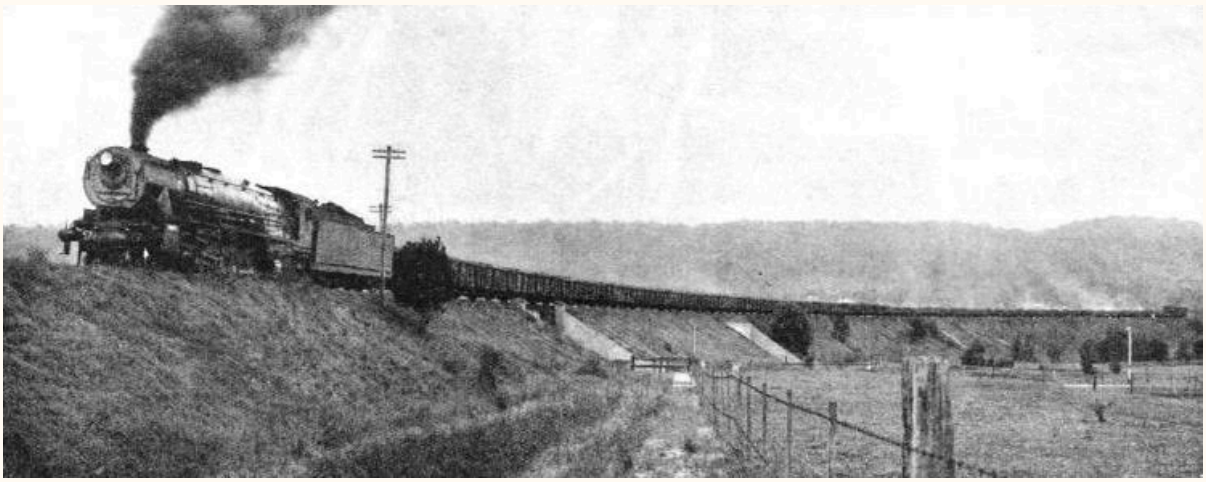
Thus the preliminary survey was a tremendous task. The Commonwealth sanctioned £20,000 for the work, and in 1908 survey parties set out from both Kalgoorlie and Port Augusta, the other from Kalgoorlie on July 1, and the one from Port Augusta in June, equipped with camels to carry food, water and gear. The Kalgoorlie party had ninety-one camels and the Port Augusta party eighty. Both worked towards the inter-state boundary on the 129th parallel.

The chief engineer of the Kalgooriie party went ahead of the main party; to the rear camel of the pack-string was attached a heavy bullock chain knotted at the free end to drag along the ground and define a trail which aided the pegging out of the grade. The chief engineer kept a compass course through the wilderness, and took stellar observations every night to check the day's work and position. By September 27 he had pegged out 455 miles - six miles a working day - and he put the junction peg on the State boundary and returned to Kalgoorlie. The party from Port Augusta, who had to plot 608 miles, reached the junction on March 13, 1909. They had averaged three miles a day, and had been delayed by water troubles.

130 Degrees in the Shade

Construction began at the Port Augusta base on September 14, 1912, and at Kalgoorlie on Feb 12, 1913. The line was scheduled to be finished in 1915, but the Great War intervened and added to the difficulties. Despite this, the rails were linked at the boundary on October 17, 1917. The first through train left Port Augusta with an official party on October 22, 1917. The feat was obscured by the war, otherwise it would have held the attention of the whole world. The gauge is 4 ft 8½-in. About 140,000 tons of rails and about 2,500,000 sleepers were used. Some 5,000,000 cubic yards of earth and rock were removed.

It was more like a campaign than plate-laying. In addition to the lack of water, there was no food or labour to be obtained on the route. First went the surveyors - camels carrying their supplies - then followed in order the clearing gangs, earthwork gangs, the track-layers, and, finally, the lifters and packers.



HAULING A TEST LOAD of 2,000 tons. A "D57" type goods locomotive in New South Wales hauling a giant freight train, which can be seen curving away into the distance.

After twenty-seven miles of track had been laid on the eastern, and twelve and a half miles on the western side, the two mechanical track-layers began biting at the thousand-miles gap.

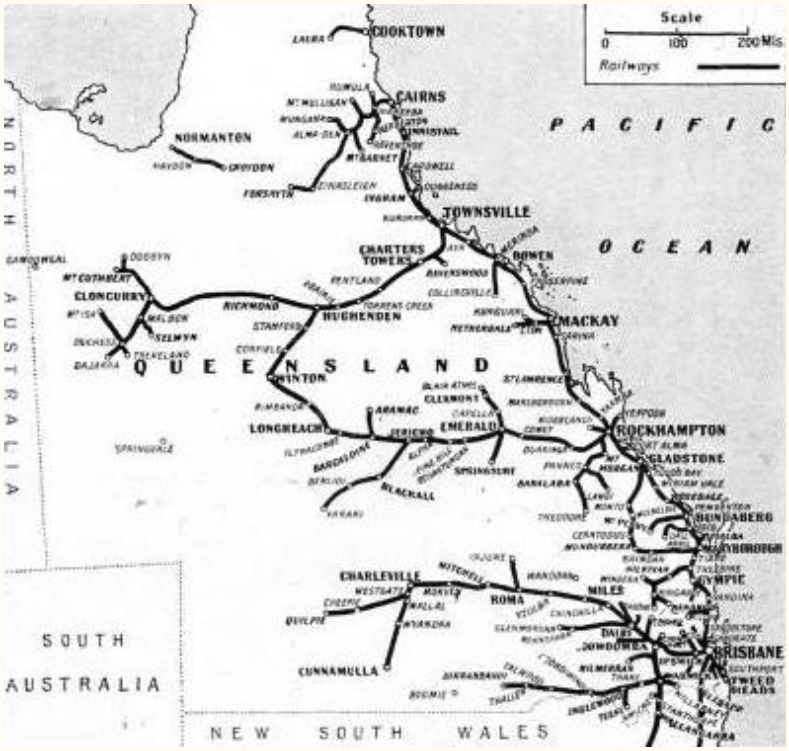
As the rails reached out into no-man's land the camps moved as often as once a fortnight. Then war broke out and skilled men were withdrawn to make munitions. Supplies of rails were stopped for long periods. Unskilled men had to learn as they worked. The original 3,000 men were considerably reduced in number.

The summer when the section of fifty miles in the sand hills was tackled proved unusually hot, and the heat of the sand hills was terrible. On one occasion a temperature of 130 degrees in the shade was recorded. The engineer-in-chief said that in the heat of the day the ink dried on his pen and it was impossible to write. The lead dropped out of shrivelled pencils. Sleepers and rails were too hot to touch. In an effort to escape from the burning sun of midday, which blistered their flesh, men buried themselves in the sand.

The navvies were paid thirteen shillings a day. No alcoholic liquor was allowed. The Government charged twenty-five shillings a week for boarding each man at the ranches, or boarding-houses. The Australian navvy, the most independent worker in the world, was well cared for. He had milk with his breakfast porridge, ate fruit and vegetables regularly, had fresh fish 800 miles from the coast, and three courses for dinner.

The organizing genius of the whole achievement was Colonel Norris G. Bell, engineer-in-chief, who preferred to remain in the background. Whenever he was asked about his great work he replied by talking enthusiastically about the work of his officers and engineers.

Cold facts and figures cannot convey the achievement of the Australians. No mere measurements and particulars can bring home the perseverance of the men who built that railway. They worked in intense heat that split their nails, pestered by hosts of flies and blood-sucking insects. The explorer, Edward John Eyre (1815-1901), the first to cross the desert in 1840, counted twenty different kinds of blood-sucking insects in four inches of his flesh at one time, each leaving an irritating, aching sting.



THE RAILWAYS OF QUEENSLAND. This map shows the chief routes in north-east Australia, which connect the interior with the ports and towns on the Pacific coast.

The men who built the trans-Australian Railway had to endure the insects and proceed with their work. They were supported by a wonderful organization with the brains behind it to understand their difficulties. The Commonwealth Railway pioneers could not go out into the desert and kill the pests before the men began work, as the Americans did before cutting the Panama Canal, because there are no pools to spray with paraffin. But they established a medical service on wheels, with doctors and dispensers. Each camp train of seven to ten coaches was protected against the heat by double roofing and mosquito- and fly-proof netting.

When the day's work was over the toilers had respite. As they progressed they were supported by the welfare organization; their children were taught in schools, and news of the outside world was brought to them.

The great feature of the railway is the Nullarbor Plain; there is nothing like it anywhere in the world. Scientists say it was once under the sea, and the fact remains that when rain falls the water sinks through the limestone into subterranean reservoirs; when it is pumped up it is too salt for locomotives. Stock can drink it, but not engines. Nullarbor is derived from the Latin (*nullhus arbor* - no tree), and is, indeed, well deserving of its grim and cheerless name.

Only dingoes and rabbits live in the holes and caves. The Australian bustard, a bird weighing up to twenty-five pounds, is seen, and at night the wail of curlew and plover is heard. In the spring there are wild ducks, swallows, wrens and finches. Eagles and smaller hawks hover, looking for rabbits. Sometimes a cormorant drifts on to this sea-like expanse of grass and bushes where there is neither surface water nor fish.

To return to the water problem: at the Western end Kalgoorlie itself depends upon a pipeline to the Mandaring Weir, 350 miles away - which is farther than from London to Land's End. At the Port Augusta end of the line there

was a limited town supply. Between the two points water had to be discovered or created.

At a point approximately 210 miles from Port Augusta the engineers were fortunate, for they tapped usable water at a mere 60 ft, obtaining as much as 120,000 gallons in twenty-four hours, and a regular 60,000 gallons a day over long periods. At other points they went as deep as 1,470 ft. The water did not gush up - it had to be pumped.

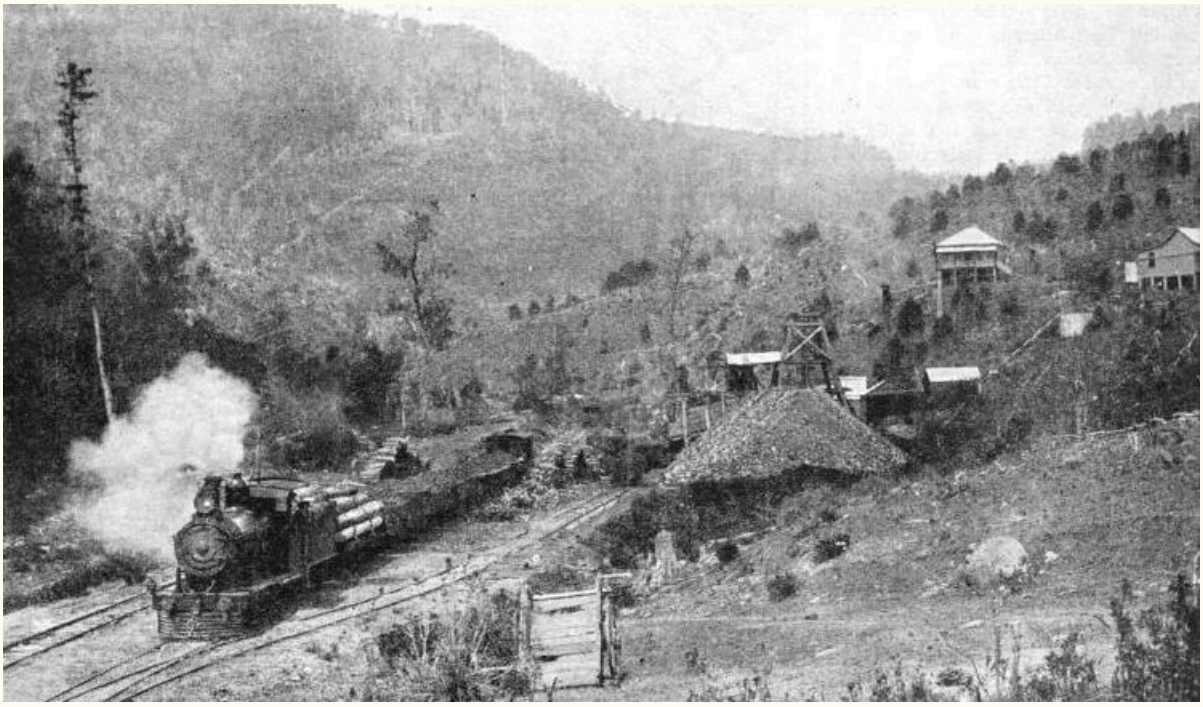
A World Record

On a stretch of nearly 150 miles every bore put down produced only salt water, and in other instances, water suitable for stock but useless for locomotives. At two points salt water was condensed for the engines, and at one of these points fuel for the condensers was hauled several hundred miles,

The precious water was guarded in dams and cement tanks which had to be roofed. It was found at first that the dry air soon caused evaporation, but roofing prevented this.

When on October 17, 1917, the two tracklayers met and the line was completed a wonderful organization had grown up in the course of the work. There were hospitals, post-offices and savings banks all on wheels. At the larger depots schools were established and there were hundreds of children along the line. In addition to locomotives over 2,500 people, nearly 500 horses and many camels had been employed.

Train speeds average twenty-eight miles an hour. The track runs without a curve for a total of 328 miles, and this is claimed as a world record for a straight stretch of line.



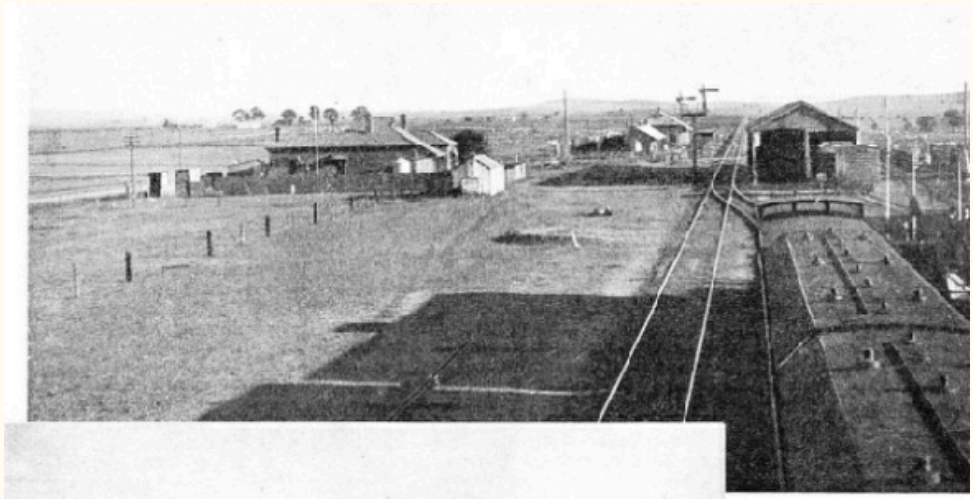
A COAL TRAIN at the Tannymorel coal mines on the Killarney branch of the main Queensland railway system.

The “Trans”, as the railway is affectionately called in Australia, has several distinctions. The Nullarbor Plain section is dead straight and almost dead level.

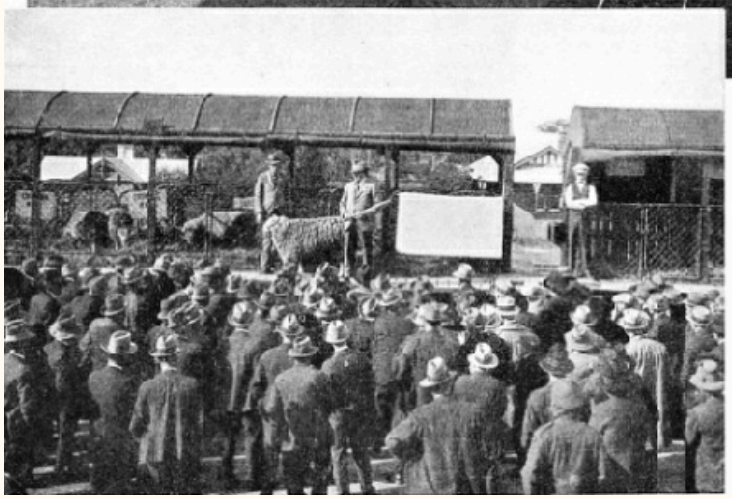
In addition, the Trans-Australian is said to be the only railway that includes the cost of meals on the train in the price of its tickets and insists that every passenger, first or second class, must book accommodation and take a sleeping-berth. The full use of the dining-car and sleeping-car is thus assured, and the hauling of empty cars across that remarkable expanse of country is avoided.

For the first 167 miles from Kalgoorlie the “Trans” passes through timbered country covered fairly thickly with the gum trees so typical of the Australian landscape; but directly the granite dips below the limestone the gum trees vanish with a suddenness so startling that a line could be drawn north and south across the track, to the west of which the gums dominate the scenery, while to the east there are none. From this spot there is not a single eucalyptus till the mallee gums of the Ooldea sandhills, 450 miles to the east, are reached.

Naretha, 205 miles from Kalgoorlie, lies to the east of a ridge with a fair growth of black oak, myall and mulga, and from this point the line runs over the Nullarbor Plain. Except for a narrow belt of black oak and myall crossing the line at 286 miles, there is not a tree, and only at long intervals a bush more than three or four feet high. Then begins the true Nullarbor Plain.



ALICE SPRINGS. The terminus of the Central Australian Railway, which line is controlled by the Commonwealth of Australia. The line runs almost to the heart of Australia, from Port Augusta, on the coast.



FARMING LESSONS FROM THE TRAIN. The official demonstration train which visits outlying districts and gives instruction to farmers

The Nullarbor is not quite dead level, but rolls away, mile after mile, in gentle undulations. A foot or so of red soil covers the limestone, but on all the rises fragments of broken limestone project upwards through the soil or lie loose upon the surface. Here and there are dongas or slight depressions in which a greater depth of soil collects; in spring these show a luxuriant growth of grass and vegetation. Most of these dongas are small, of a few acres, but some near the western end of the Nullarbor Plain contain hundreds of acres. Some of the dongas are covered with grass a foot long, waving in the wind, and dotted with pink and white daisies and other flowers. Dongas are rare in the great central area but reappear towards the eastern edge of the plain.

Characteristic plants of the plain are the bluebush, with its ghostly colouring; and the saltbush, best of all native fodder plants, with its greyish-green leaves, salt but not unpleasant to the taste. Alone, or with a few kindred plants, they cover almost the whole surface.

Grass, which comes in the spring, withers before the heat of summer, but the saltbush and the bluebush are perennial. These bushes are the great feature of the plain. Their leaves absorb the moisture from every chance shower, and when there is no rain they absorb the dew, and so survive when the grass withers. Here and there a tiny dwarf acacia raises spine-like leaves and yellow fluffy balls a few inches from the ground, but it seems to be a stranger that has strayed into an uncongenial environment.

There is a great fascination about the Nullarbor. The single track of the railway stretches ahead and behind without a curve, the shining rails running on into infinity, seeming to meet in the distance. There is nothing but plain and sky.

The Nullarbor to the north of the line is unexplored country, although it has been conjectured that the plain has a total area of nearly 100,000 square miles.

Beneath the limestone of the Nullarbor are caverns and passages. Blowholes are numberless, many blind and shallow, others leading to the strange underworld. The blowholes are the lurking places of centipedes, spiders and beetles, while cave-owls make their homes in the ledges and crevices, as there are no other bird-homes on the treeless plain.

The line runs across the border into South Australia, 452 miles east of Kalgoorlie, where a stone cairn marks the boundary: the great plain stretches for another 150 miles, until the Musgrave Ranges come into view to the northeast. Then a long “edge” runs north and south across the plain.

Annual Rainfall - 4 Inches

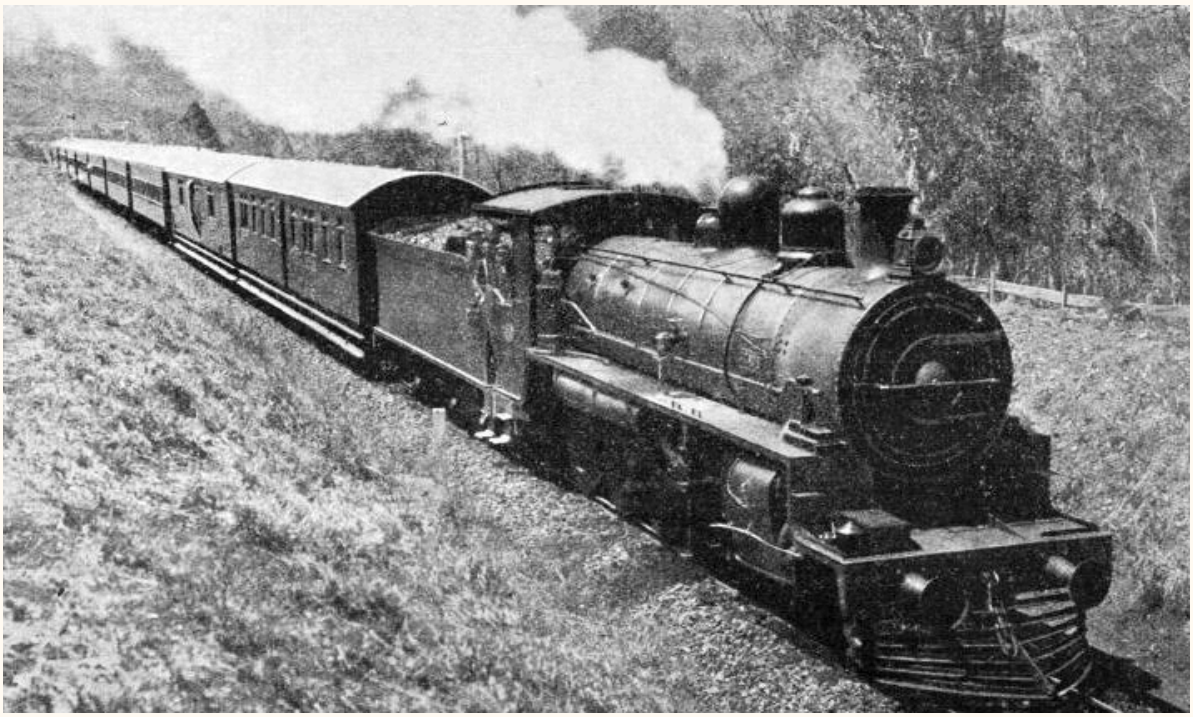
The line dips about eighty feet, and the character of the country begins to change. Real trees appear, only ten or twelve feet high, though compared with the bushes of the plain they are giants. Grass grows in the hollows. A few

miles more and this country shades off into the sandhills. The Nullarbor Plain is left behind.

At Ooldea, 624 miles from Kalgoorlie, the sandhill belt begins. It is a tangle of ridges and hollows of a fine dust which looks like sand. The ridges and hollows are covered by small trees, but when the surface is cleared the soil is easily moved by the wind.

The last division of the line, from the sandhills to Port Augusta, is more varied than any of the others. For about a hundred miles, until the hilly country round Tarcoola is reached, the line runs across red-soil plains and undulating country. At Wynbring, 321 miles from Port Augusta, the granite, which has been hidden for 560 miles, comes to the surface again.

Fifty miles farther is the first fence the line has crossed since leaving Kalgoorlie, the outer boundary of the Wilgena run, which is eighty miles long and forty miles wide. On a corner of it is the Tarcoola goldfield, where the rocks come to the surface and form low, rugged hills. The hills die away again as the “Trans” reaches the “Lake Country”.



THE MAIL TRAIN FROM SYDNEY. This train proceeds from the capital of New South Wales to Brisbane, the capital of Queensland, a journey of 613 miles, by way of Newcastle, Gloucester and Grafton.

The lakes are vast, shallow pans, some of them hundreds of miles square, covered with a shallow sheet of water after rains, and becoming mere beds of salt in the dry season. But the shallow water is as blue as the sea. and the lakes are picturesque enough with their waters reflecting the sun, and surrounded by bluffs, and mimic cliffs, with wooded slopes beyond. The line crosses a corner of Lake Hart, and later it skirts the shores of the Island Lagoon, which has a remarkable hill, peaked like a miniature volcano, standing in the middle of it. The line crosses Lake Windabout, beyond which is Pernatty Lagoon, the waters of which are impregnated with copper.

After fifty miles or more of plain country, with strange flat-topped hills rising to the south-west - remnants of a once continuous table-land - the line crosses the extreme head of Spencer Gulf and reaches its eastern terminus at Port Augusta.

No part of the long stretch of country has an average rainfall of more than 10-in a year, while the average for the whole is probably less than 4-in. Provision for water storage for about 300 miles at each end of the line to the extent of about 53,000,000 gallons has now been provided for railway purposes.

The regular passenger service is two trains a week each way, but many special passenger trains are run. Out of the total coaching stock half are sleeping cars, and it is unlikely that any other railway in the world could show such a high per-centage. Owing to the break of gauge at each end of the line, there is little freight traffic. More than twenty-five per cent of the service vehicles are water tanks.

The length of the “Trans” express depends upon the number of bookings. There are usually seven corridor coaches, including dining-car and lounge-car. The first-class sleeping-cars have two-berth compartments, the second-class having four-berth compartments. The lounge car is divided into smoking and nonsmoking saloons, the latter having a piano. The trains are fitted throughout with automatic couplings, and increase speed and stop very smoothly, while they ride easily.

The electric staff system of working is in operation throughout, and each station is also a telegraph depot. Every gang of workmen not located near a telegraph station, and every train are equipped with portable telephones and appliances for making connexion with the electric staff wire at any point. The staff sections are widely spaced, and the portable telephone is therefore an important precaution in the event of a break-down or of any other emergency.

News is telegraphed to points along the route and made available to passengers.

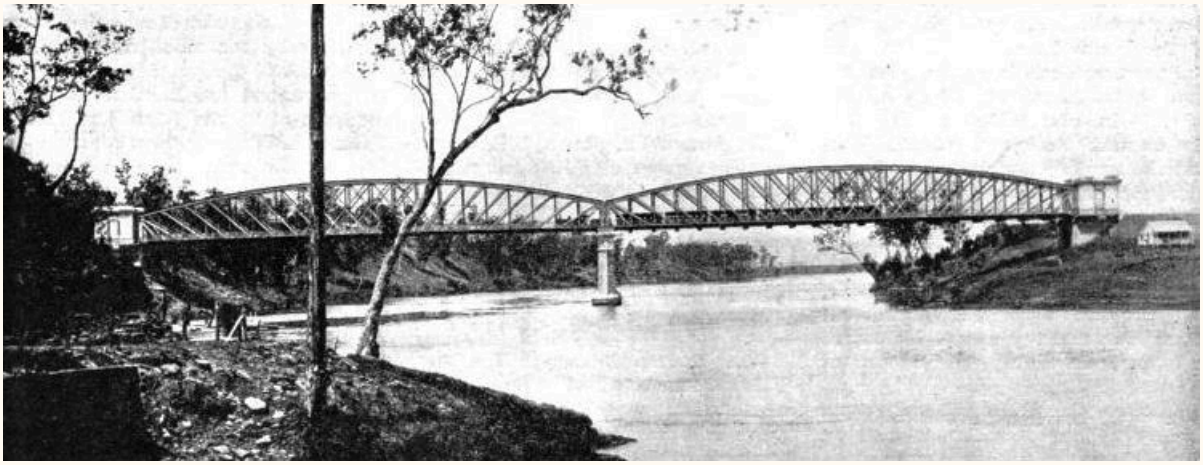
Change of Locomotives

Two splendidly appointed special cars are available. Car No. 1 is a self-contained car accommodating up to eight persons. There are two bedrooms and three two-berth compartments, dining-saloon, toilet, bath-room with hot and cold water, a kitchen, electric heat and light, and an observation platform. This car was used by the Prince of Wales. Car No. 2 is not quite so elaborately appointed.

The change of engines in the run from Port Augusta is made at Cook, 513 miles from Port Augusta and 538 miles from Kalgoorlie. The locomotives are of the 4-6-0 type passenger engine, of the “P” class, and the average load of the Transcontinental trains is 420 tons.

Loads up to 470 tons, however, have been worked across the Nullarbor Plain - a tonnage considerably in excess of the average passenger train loads in Australia. The two runs of 513 miles and 538 miles are the longest continuous locomotive workings in any part of Australia, and are longer than any through locomotive workings in Great Britain, though the speed of this Australian service, of course, is not high.

In every part of the line the prices of the goods and food are the same. The supply train is not only a retail shop, but is also the great event in the routine of the workers in the wilderness. Supplies of bread are sent twice a week, and a butcher on the train kills sheep while the van is travelling, and so times his work that by the time the train arrives at each station the meat is ready for sale. There is a dispensary van, and the whole organization is a credit to the Commonwealth Government Railways, who have spent nearly £8,000,000 to connect the Western State Capital with her sisters. The time saved between Melbourne and Perth is about three days, as compared with sea transport.



A GRACEFUL DOUBLE-SPAN. The Indooroopilly Bridge near Brisbane, Queensland, is a fine example of railway-bridge construction in Australia.

It must not be thought that the title, Trans-Australian Railways, means that the Commonwealth Government Railways are responsible for the passenger from the time he entrains at Perth until he reaches Brisbane. Each State owns the section of the line within its borders.

After having booked in advance we board the train at Perth, bound for Brisbane. (The fare from Fremantle, the port for Perth, which is twelve miles away, is the same.) We have paid £23 14s 6d for our first-class ticket, which provides reserved seats, sleeping-berths and meals on the Perth-Kalgoorlie and Trans-Australian lines; after Port Augusta we will have to pay for our meals and also for sleeping-berths. First we are on a 3 ft 6-in gauge of the Western Australian Government Railways. At Kalgoorlie we have to change trains for the 4 ft 8½-in standard gauge of the Trans-Australian Railway; at Port Augusta we change to the 3 ft 6-in of the South Australian Government Railways, at Terowie we change to the 5 ft 3-in gauge, at Albury to the 4 ft 8½-in of the New South Wales Government Railways. South Australia’s capital, Adelaide, and the Victorian capital of Melbourne were, until recently, the only two capitals linked by railways of the same gauge. The Queensland gauge is 3 ft 6-in. We have also to change trains at Adelaide, Melbourne and Sydney, having to wait seven hours and a quarter at Melbourne and nine hours twenty-eight minutes at Sydney.

Therefore the reader will understand why Australians are apt to make strong comments on the subject of gauges.

Now that we are in Brisbane we can see how varied are the problems that confront the Australian railroad engineer. The builders of the track across the Nullarbor Plain did not have enough water, the Queensland engineers have too much of it. Innisfail has an annual rainfall of 150-in. This place is some thirty miles south of Cairns, which is seventeen degrees south of the equator, and the rains are of the monsoon character.

From the beginning of the first railway Queensland pursued an active policy of decentralization, and lateral lines were built long before any coastal railway was undertaken. The three main systems, southern, central and northern tap all the pastoral lands of the State.

The Queenslander is the most light-hearted of all Australasians. Sir George Bowen, the first Governor, said that when he took office he found nothing in the treasury except a few coppers. The new State soon had a better bank balance. In 1863 the Premier, the Hon. R. G. W. Herbert, visited England, and as the result of his observations the first railway Bill was introduced. Ipswich was decided upon as the starting point for the railway, tenders were invited and the contract for a twenty-one and a half mile railway to Grandchester was awarded to the English firm of Peto, Brassey and Betts. Construction work began on February 25, 1864. Six months later a vessel arrived from England with rolling-stock and a number of mechanics. The first section was opened on July 31, 1865. In those days there were two Houses of Parliament; both adjourned for a fortnight to attend the opening ceremony at Grandchester and a racing carnival at Toowoomba.

Mr. Abram Fitzgibbon, C.E., recommended a gauge of 3 ft 6-in, and his advice was followed. The policy has been criticized, but the Queenslanders maintain that it is the best for their State, and has never once proved inadequate.

Jungle Clearing

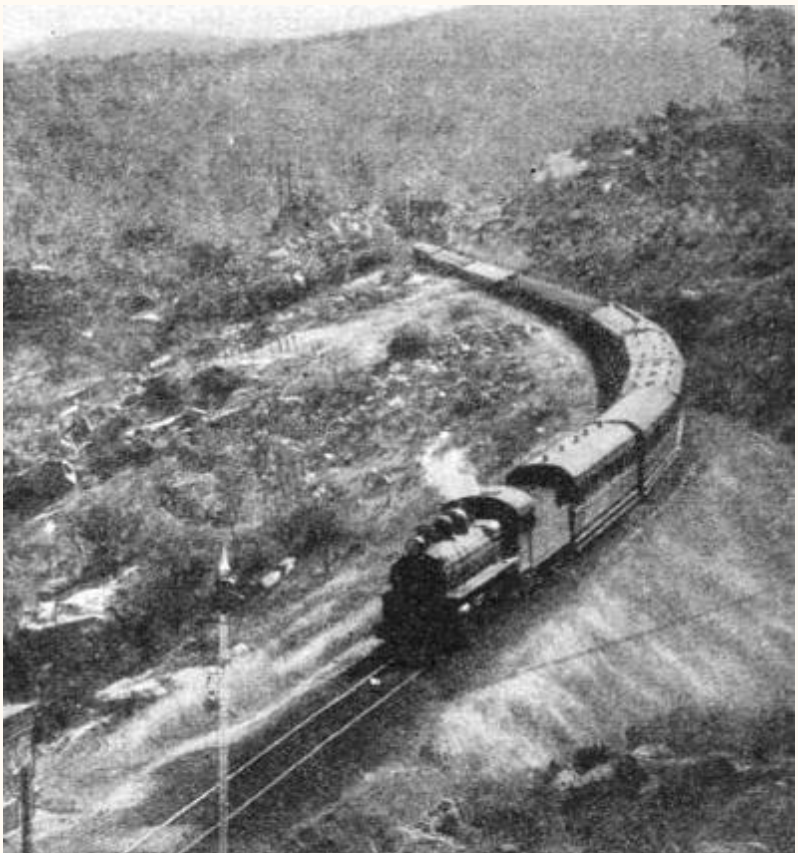
When their neighbours in New South Wales (gauge 4 ft 8½-in) make comments, they reply that the smaller gauge is better for Queensland, with her sparse population and large country.

“We have never heard of New South Wales desiring to send us any goods which could not be carried by Queensland”, Queensland declares. There are heavy ranges with sharp curves on the Queensland side of the border. The most difficult item for the Queensland railways to carry was a German tank, which had been captured by Australian troops in France.

The difference in gauges formerly necessitated transhipment of passengers and goods at border stations, but this difficulty has been overcome by the construction of a 4 ft 8½-in gauge line from Brisbane to Kyogle, New South Wales. This shortened the former journey by 102 miles, and eliminated the break in the gauge.

Queensland has the largest railway mileage of all the Australian systems. The main line to the north is more than 1,200 miles long, and there are lines running into the interior nearly 600 miles long. No other State in the Commonwealth has so many dead-end branch lines in proportion to the total mileage. Rail motors are being used more extensively on the branch lines.

It is during the rainy season in the mountains that the engineers have to be on the alert. After a storm an engineer on such a line may find that a bridge has been swept away by the tremendous volume of flood water; a torrent has taken the track in its stride, a landslide has left a stretch of line in the air, or huge trees have crashed upon the track. It is then that the vehicles fitted as breakdown wagons and the travelling crane come into use. There are wagons fitted with appliances to spray poison on the track, for not only does the railway engineer have to battle with storm, he must also keep the way through the jungle clear. Trains on some lines are so infrequent that the jungle is swift to close impenetrably upon the track. These lines depend upon seasonal traffic, and may be idle for a long period.



THE KALGOORLIE-PERTH express on its 375 miles run. This picture shows the train crossing the Darling Ranges in Western Australia.



The railway looks after its passengers' needs, and the charge for meals is very reasonable. At the central station the charge for breakfast is 2s. (It should be explained that Britons in the tropics call breakfast a meal eaten between 11am and 2pm.) A three-course luncheon or dinner is priced 1s 6d.

Passengers may obtain on application to the station-masters, an ice-chest, the charge for the hire of which is 1s 3d for the first 100 miles and 5d for every additional fifty miles or part thereof. The chest is of convenient size for placing under the seat of the carriage, and is of great convenience to travellers during the summer months.

South of Queensland the two great States of New South Wales and Victoria have developed one of the finest railway systems in the Southern Hemisphere.

You can read more on "Australian Engines", "Travel in Australia" and "Electrification Overseas" on this website.

You can read more on the Trans-Australian Railway in **Wonders of World Engineering**

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